Corbett maths

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Examples

Workout

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Question 1: Simplify each of the following

(a) $y + y + y + y = 4y$ (b) $w + w + w + w + w = 5w$ (c) $a + a + a + a + a + a = 6a$ (d) $s + s + s = 3s$

(e) $n + n = 2n$ (f) $g + g + g + g - g = 3g$ (g) $y + y + y + y - y - y = 2y$ (h) $p + p - p - p = 0$

(i) $3y + 2y = 5y$ (j) $4a + 3a = 7a$ (k) $9k + 5k = 14k$ (l) $7m + m = 8m$

(m) $15c + 20c = 25c$ (n) $6w - 3w = 3w$ (o) $10y + 3y - 5y = 8y$ (p) $20t - 14t = 6t$

(q) $7x - 3x - x = 3x$ (r) $8k - 8k = 0$ (s) $7y - 2y + y = 6y$ (t) $5u - 4u = u$

(u) $y^2 + y^2 = 2y^2$ (v) $a^2 + a^2 + a^2 = 3a^2$ (w) $c^2 + c^2 + c^2 + c^2 + c^2 = 5c^2$ (x) $7y^2 + 3y^2 = 10y^2$

(y) $2w^2 + 4w^2 + 8w^2 = 14w^2$ (z) $6y^2 - 2y^2 + 3y^2 = 7y^2$

Question 2: Simplify the following expressions

(a) $4u - 6u = -2u$ (b) $8w - 9w = -1w$ (c) $4a + 2a - 9a = -3a$ (d) $2y - 9y = -7y$

(e) $-3g - 2g = -5g$ (f) $-4f + 9f = 5f$ (g) $-m - 7m = -8m$ (h) $5y^2 - 7y^2 = -2y^2$

(i) $6a^2 + 2a^2 - 9a^2 =$ (j) $ab + ab + ab = 3ab$

$8a^2 - 9a^2 = -1a^2 = -a^2$

Question 3: Simplify the following expressions

(a) $3a + 2b + 4a + b = 7a + 3b$ (b) $7y + 5y + 2h + 2h = 12y + 4h$ (c) $g + 8a + 2a + g = 2g + 10a$

(d) $7m + 7p + 8m + p + 2p =$ (e) $9e + 2 + e + 2 = 10e + 4$ (f) $4 + 3a + 2a + 8 = 5a + 12$

$15m + 8p + 2p = 15m + 10p$

(g) $2y + 4 + 3y - 1 = 5y + 3$ (h) $8 + 3w - w - 3 = 8 + 2w - 3$ (i) $5 - 4s - 2 + 10s = 6s + 3$

(j) $3x + 6y + 5x - 2y = 8x + 4y$ (k) $6m - 2s + 11s + m$ (m) (l) $2a + 3b - 2 + a + 3b + 4 = 3a + 6b + 2$

$7m + 9s$

(m) $3a - 2b + a - 5b = 4a - 3b$ (n) $2x - 2y - 6x + 5y = 0$ (o) $y - 4m - 3y - 5m = 2y - 1m$

$4x + 3y$

(p) $7p - 2q - q + 3r + 4r = 0$ (q) $11c + 8d - 6c - 11d = 5c + -3d$

$7p - q + 7r$

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Question 4: Simplify the following

(a) $3y^2 + 4ab + 7y^2 + ab = 10y^2 + 5ab$ (b) $9x^2 - 2x - 11x^2 + 5x = -2x^2 + 3x$ (c) $7ac - 3ab + 9ab - 7ac = 6ab$

Question 5: Expand and simplify the following

(a) $2(y + 3) + 3(y + 1) = 2y + 6 + 3y + 3 = 5y + 9$ (b) $8(x + 2) + 3(x + 3) = 8x + 16 + 3x + 9 = 11x + 25$ (c) $4(x - 1) + 2(x + 3) = 4x - 4 + 2x + 6 = 6x + 2$

(d) $5x + 3 + 2(x + 9) = 5x + 3 + 2x + 18 = 7x + 21$ (e) $3(2y + 1) + 4(2y + 5) = 6y + 3 + 8y + 20 = 14y + 23$ (f) $5(2x + 3) + 2(3x + 1) = 10x + 15 + 6x + 2 = 16x + 17$

(g) $7(c + 2) + 3(c - 2) = 7c + 14 + 3c - 6 = 10c + 8$ (h) $5(2a + 7) + 2(9a - 4) = 10a + 35 + 18a - 8 = 28a + 27$ (i) $9(t + 3) + 3(2t - 11) = 9t + 27 + 6t - 33 = 15t - 6$

(j) $2(x - 4) + 5(x - 2) = 2x + 8 + 5x - 10 = 7x - 2$ (k) $6(y - 1) - 2(y + 3) = 6y - 6 - 2y + 6 = 4y - 12$ (l) $8(x + 2) - 3(x - 2) = 8x + 16 - 3x - 6 = 5x + 10$

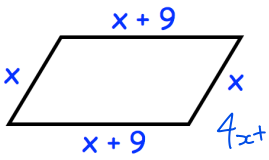
(m) $5(2y - 3) + 3(y - 2) = 10y - 15 + 3y - 6 = 13y - 21$ (n) $2(4w - 5) - 2(w - 7) = 8w - 10 - 2w + 14 = 6w + 4$ (o) $5(3y + 7) - 3(2y - 5) = 15y + 35 - 6y - 15 = 9y + 20$

Apply

$$\begin{array}{r} 2\ 13 \\ -27 \\ \hline 06 \end{array}$$

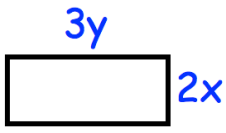
Question 1: Write down the perimeter of each shape below

(a)



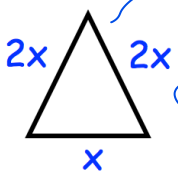
$4x + 18$

(b)



$4x + 6y$

(c)



Answers.

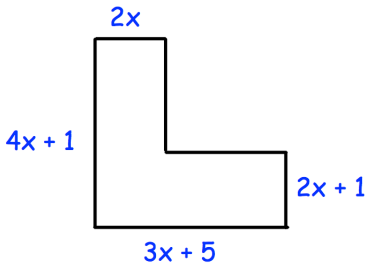
$5x$

Question 2: A square has a side length of 3x.
Find an expression for the perimeter of the square. $3x \times 4 = 12x$

Question 3: $6x + 7y + x - 8y = 7x - y$
Write down three other expressions that are equal to $7x - y$: $6x + 1x - y$, $7x + y - 2y$, $2x + 5x - (5y - y)$

Question 4: Find an expression for the perimeter of this shape

$(2x) + (4x + 1) + (3x + 5) + (2x + 1) = 11x + 7$



Answers

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